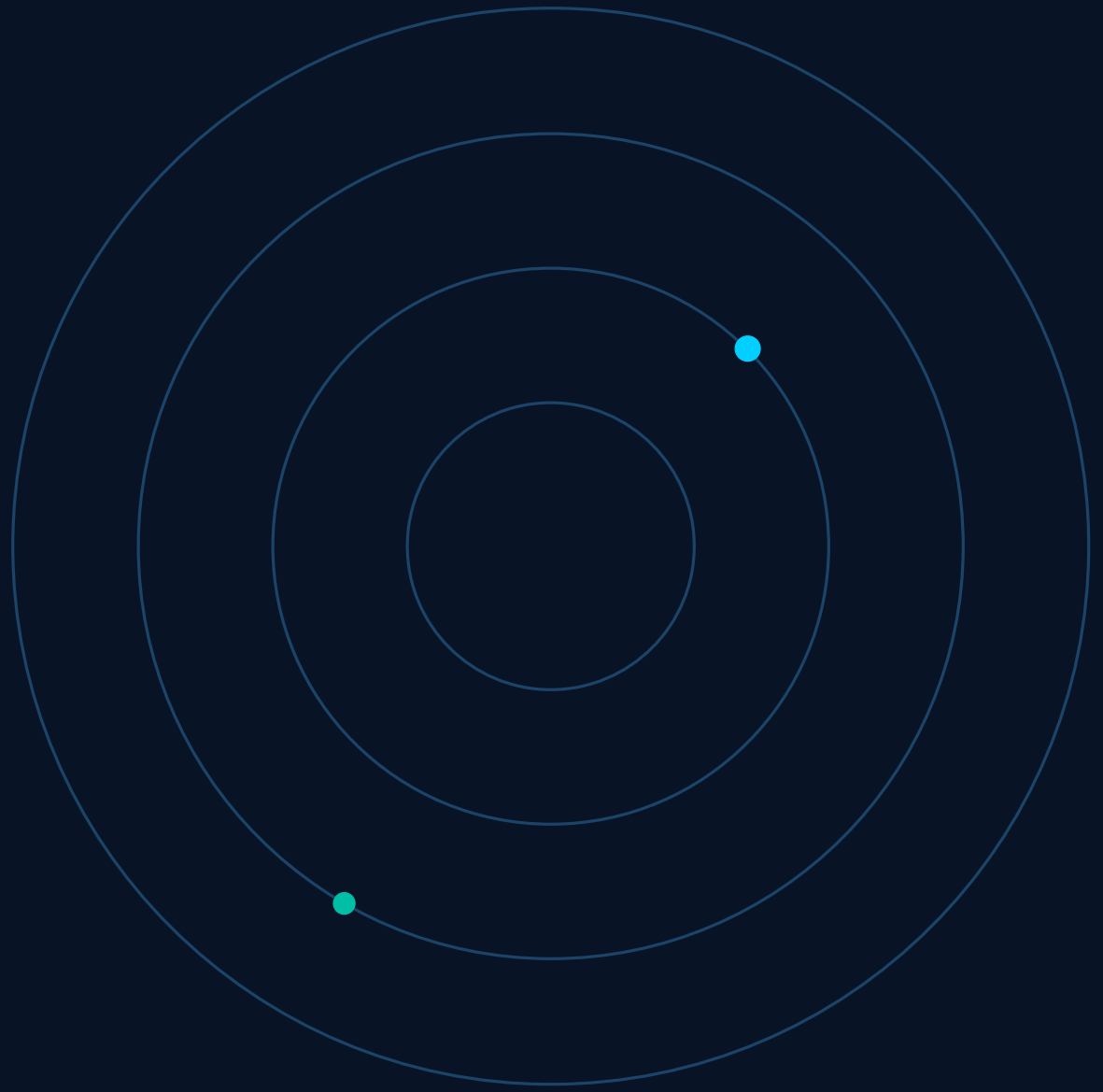
PRAVHA

Intelligent Rainwater Harvesting & Adaptive Stormwater Management

Every Drop Should Have a Purpose.

Falling Sun 2026 · Robotics



Rainfall Is a Resource. Unmanaged Runoff Is a Challenge.

1

WATER QUALITY

Initial rooftop runoff may contain accumulated contaminants.

2

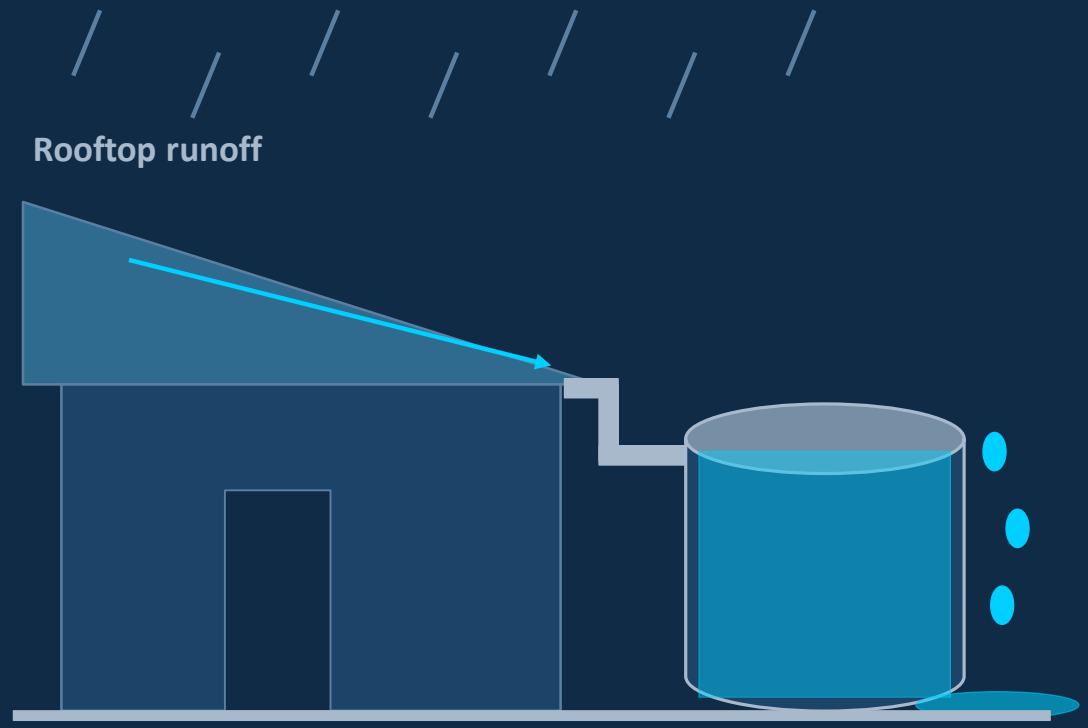
STORAGE

Storage tanks can overflow without coordinated diversion.

3

OVERFLOW DESTINATION

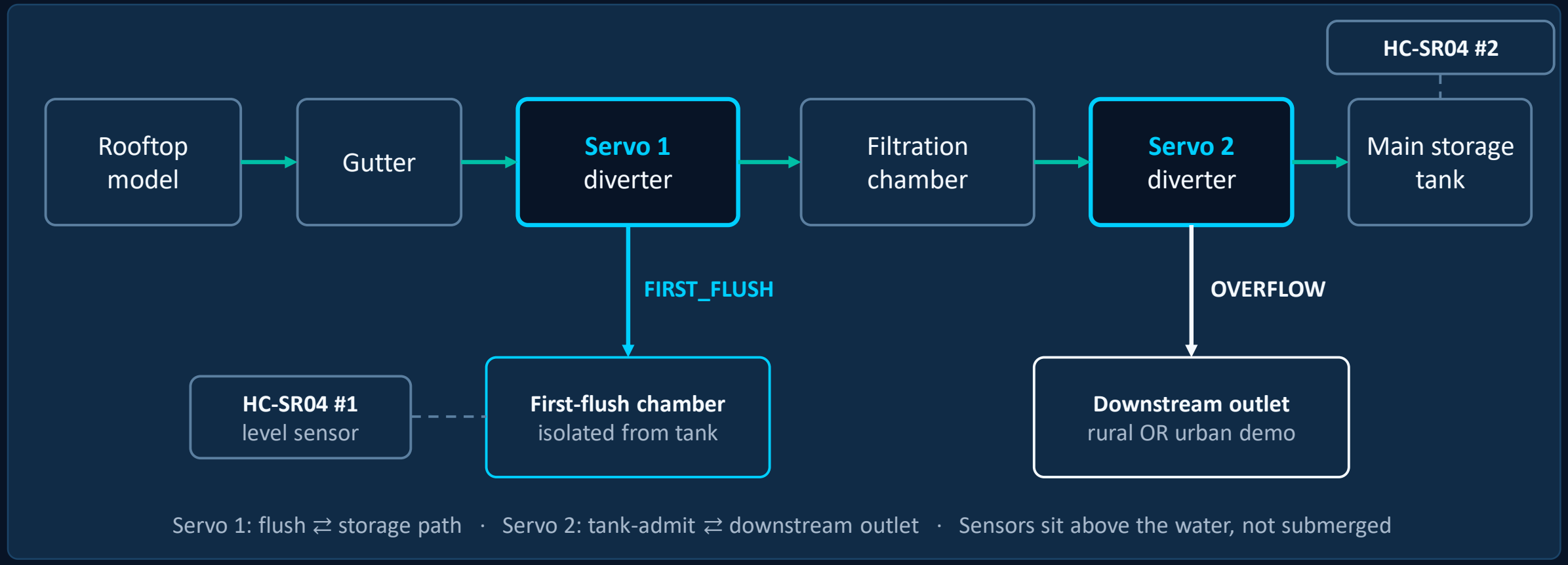
Urban drainage systems face runoff pressure, while suitable rural sites offer infiltration opportunities.



Conceptual illustration not a PRAVHA prototype

One Intelligent System. Complete Rainwater Management.

Rain Detection → Rooftop Collection → First-Flush Separation → Filtration → Storage → Overflow Management



RURAL suitable infiltration / Groundwater recharge

URBAN sponge-garden / rain-garden demonstration

The Intelligence Behind PRAVHA

CONTROLLER

- Arduino Uno

SENSORS

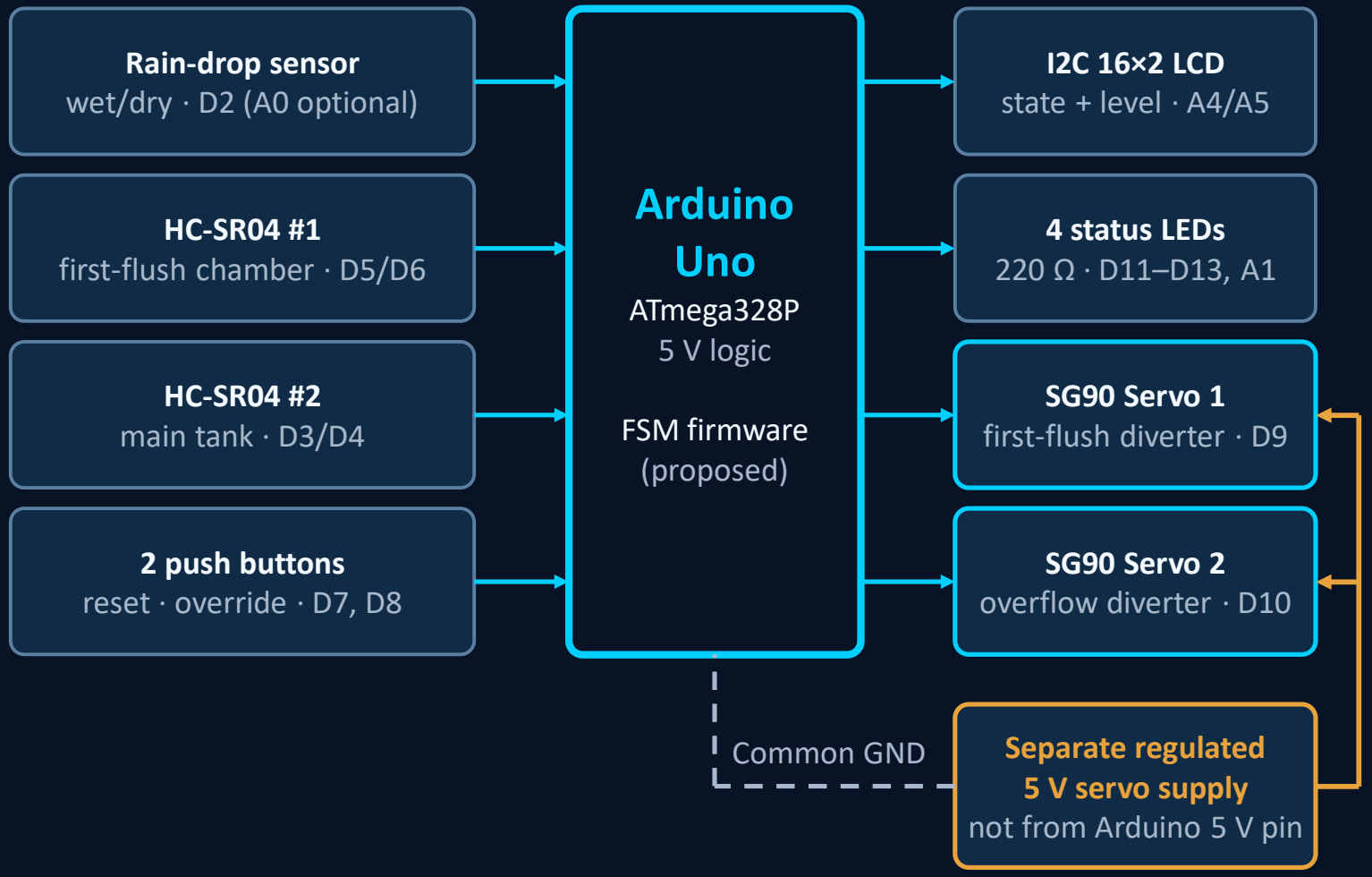
- One rain-drop sensor
- Two HC-SR04 ultrasonic sensors

ACTUATORS

- Two SG90 servo motors

USER INTERFACE

- One 16×2 I2C LCD
- Two push buttons
- Four LEDs



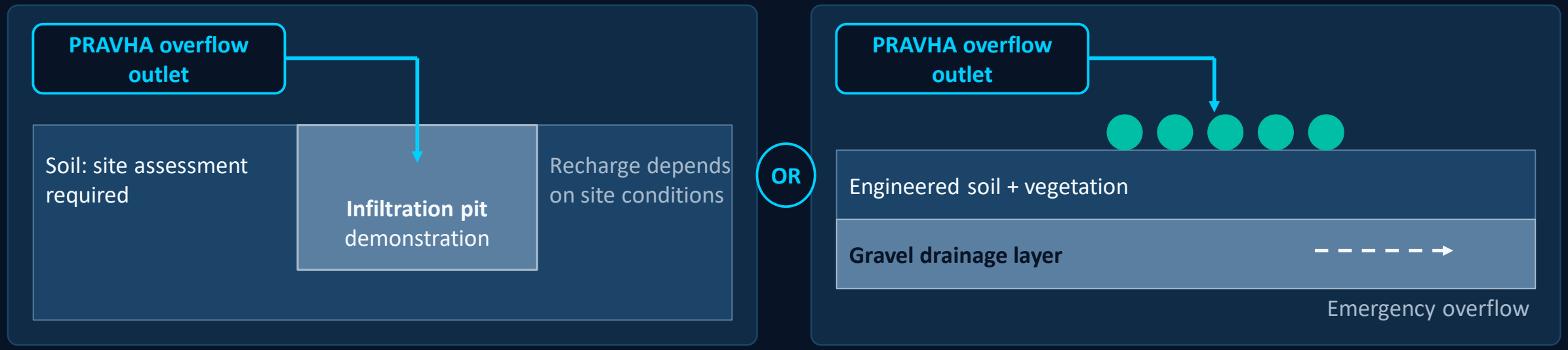
One Architecture. Two Environmental Applications

RURAL CONFIGURATION

- Rooftop harvesting
- Non-potable storage
- Controlled overflow
- Suitable infiltration/Groundwater recharge
- Potential groundwater replenishment where conditions permit

URBAN CONFIGURATION

- Rooftop runoff collection
- Automated storage management
- Controlled overflow
- Sponge garden or rain garden
- Potential reduction in unmanaged local runoff



Conceptual Deployment Configurations — Not to Scale

What Makes PRAVHA Different?

Basic rainwater harvesting	PRAVHA
Primarily passive collection	Rainfall-triggered automation
Conventional first-flush arrangements	Sensor-monitored first flush
Limited coordinated monitoring	Automatic storage monitoring
Overflow may be discharged without active routing	Servo-controlled overflow diversion
Destination not coordinated with storage state	Configurable downstream application
No integrated status indication	Visible operating status (LCD + LEDs)

Our innovation is the integration of established technologies into a coordinated, configurable, low-cost robotics architecture.

We do not claim to have invented first-flush diversion, ultrasonic sensing, servo actuation or filtration.

PRAVHA — Smarter Rainwater Management

POTENTIAL IMPACT

- Improved rooftop rainwater management
- Rural infiltration opportunities
- Automated first-flush and overflow control
- Urban stormwater retention and controlled drainage

SDG 6 Clean Water and Sanitation · **SDG 11** Sustainable Cities and Communities · **SDG 13** Climate Action

"PRAVHA does not create more rainfall. It proposes a smarter way to manage the water we already receive."

THANK YOU — QUESTIONS?